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**IMPLICATIONS OF TOPOLOGICAL COMPLEXITY
AND HAMILTONIAN CHAOS IN THE EDGE MAGNETIC
FIELD OF TOROIDAL FUSION PLASMAS**

by
T.E. EVANS

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The edge region of magnetically confined toroidal fusion plasmas, such as those found in tokamaks and stellarators, is both dynamically active and topologically complex. The topological properties of the magnetic structures observed in the active edge region of high performance poloidally diverted plasmas are qualitatively consistent with those of a time varying web of intersecting homoclinic tangles defined by invariant manifolds of the primary separatrix of the system interacting with the invariant manifolds of resonant helical magnetic islands. Here, intersections of stable and unstable manifolds produce Hamiltonian chaos in the edge magnetic field that can strongly affect the transport and stability properties of the plasma. A quantitative description of the dynamics involved in these processes requires developing a better understanding of the plasma response to such complex topologies. A review of the recent experimental observations and progress on 3D fluid, kinetic and extended magnetohydrodynamic modeling of the edge plasma in tokamaks is given in this paper. These are related to the topological structure of the invariant manifolds and the chaotic structure of the field lines based on conservative dynamical system theory.

1. Introduction

1.1. General Background

Magnetically confined toroidal plasmas, such as those found in tokamaks and stellarators, are particularly attractive for fusion energy research because of their relatively well established potential for achieving stable, steady-state, operating conditions that combine high plasma pressures with long energy confinement times and the relatively high energy densities needed to produce self-driven burning discharges. On the other hand, the edge plasma in these devices is highly susceptible to the effects of magnetic perturbations from small

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asymmetries in the external confinement and shaping coils. The edge plasma can also be resonantly perturbed by helical magnetic fields generated by instabilities deep inside the core plasma and by non-axisymmetric fields from external magnetic coils used to control these instabilities. Each of the perturbation fields can result in helical magnetic islands that overlap to produce stochastic field line trajectories that escape from the confinement region. These open field lines connect high temperature plasma particles to material surfaces that are not intended to be exposed to such plasma conditions. Alternatively, resonant magnetic perturbation (RMP) coils can be designed to control the structure of the edge magnetic field and thus to manage the plasma properties in a thin boundary layer region of the plasma called the pedestal where steep pressure gradients couple to the plasma current causing instabilities known as edge localized modes (ELMs).

Managing the power and particle exhaust from the edge of a burning fusion plasma is a formidable task in any magnetic confinement device even under ideal conditions. This is due to technological constraints imposed by the properties of high heat flux materials used in exhaust components and because of uncertainties in some aspects of the basic plasma physics operating in the boundary of these devices. In particular, the complex nonlinear nature of the interactions between the boundary layer plasma and the edge magnetic field is not well understood making this a challenging area of research. Thus, the goal of this review is to discuss recent progress made in understanding the complex relationship between structure of the edge magnetic field and the physics of plasmas that reside in the boundary region of a tokamak. Since this is an important topic, there is much more published material than can be covered in this brief review. Some of the earlier work done on stochastic boundary layer physics in circular tokamaks is covered in a 1996 review [1]. Here, the focus is on describing the implications of using a nonlinear dynamical systems approach to study magnetic field line chaos and tangled plasmas in tokamaks especially with respect to the confinement of the plasma, steady-state power exhaust issues and transient energy bursts in fusion relevant plasmas.

1.2. Steady-state Power Exhaust in Fusion Plasmas

Deuterium-tritium (DT) fusion plasma discharges in ITER [2], the first net power producing experimental tokamak reactor of its kind, are expected to last up to 400 s with inductively driven plasma currents of 15 MA and fusion

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thermal output power levels approaching 500 MW. These plasmas will be sustained with external heating systems capable of delivering injected power levels of up to 73 MW for the duration of the discharge. Thus, in ITER the fusion power gain Q , defined as the ratio of the DT fusion output power to that of the external heating power, is expected to be 10 [2] with 50 MW of injected heating power. At this operating point, the steady-state power exhausted from the plasma along open field lines intersecting the material surfaces of the exhaust system, known as the divertor in ITER, is of order 80 MW. At least 50% of this power must be radiated to the plasma facing walls before reaching the divertor surfaces in order to remain below the maximum heat flux limit of 10 MW/m^2 set by the materials used on the divertor surfaces. On the other hand, an economical fusion power generating tokamak, with DT plasmas of approximately same size as those in ITER, is projected to have thermal output power levels range between 2.5 and 3.6 GW and will require steady-state power exhaust capabilities of between 0.4 to 1.2 GW [3] or approximately $100\text{--}300 \text{ mW/m}^2$ on the divertors assuming an axisymmetric loading distribution and radiative cooling along the open magnetic field lines.

Figure 1 shows a 3D cutaway representation of the DIII-D [4] device. DIII-D is a poloidally-diverted tokamak [5] similar to ITER, with a plasma volume that is approximately a factor of 45 smaller than the plasma volume anticipated in ITER. Here, it is noted that major radius (R_0) of the toroidal plasma in ITER is $R_0 = 6.2 \text{ m}$ and the minor (poloidal) radius (a) is $a = 2.0 \text{ m}$. Thus, ITER is referred to a 6 m class tokamak with an aspect ratio $A = R_0/a = 3.1$ while DIII-D is a 1.5 m class tokamak with $A=3.0$.

Assuming the power flow out of the ITER plasma and the radiation on the open field lines is axisymmetric with respect to the toroidal angle, it should be possible to safely exhaust the steady-state heat flux produced by the plasma using open field line radiative processes. On the other hand, research in the current generation of 1.5 m class tokamaks has shown that non-ideal effects such as small resonant and non-resonant magnetic perturbation from external field coils and internal magnetohydrodynamic (MHD) modes significantly alter the edge magnetic topology producing relatively complex 3D structures. For example, in the first poloidally diverted tokamak (ASDEX), a 1.5 m class tokamak with $A=4.1$, the integrated power flux hitting the divertor target plates was shown to be toroidally asymmetric under all operating conditions. The

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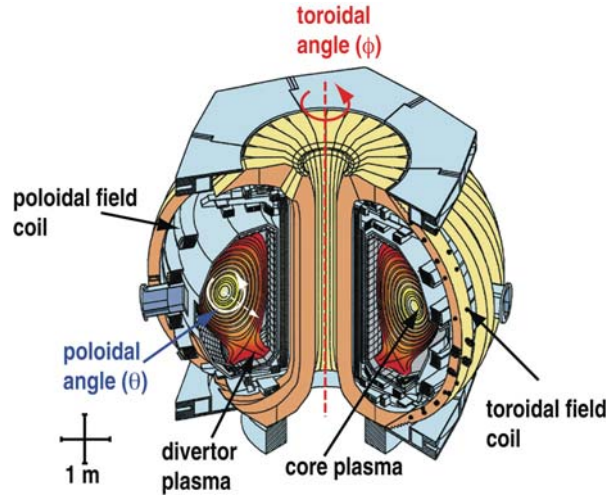


Figure 1. Cutaway 3D view of the DIII-D device, a 1.5 m class $A=3.0$ tokamak, showing the arrangement of the toroidal and poloidal field coils outside the vacuum vessel that contains a poloidally diverted plasma consisting of a core and divertor region.

magnitude of the toroidal energy deposition asymmetry in ASDEX ranged from 9 kJ/toroidal-segment at one toroidal angle to 14 kJ/toroidal-segment at a toroidal angle on the opposite side of the machine (an asymmetry factor of 1.6) during low power ohmic and ion cyclotron heated plasmas. This asymmetry factor increased to 4.5 in lower hybrid heated plasmas (30 kJ/toroidal-segment to 135 kJ/toroidal-segment) [6]. In addition, the toroidal distribution of these asymmetries was strongly affected by changes in the edge magnetic topology that were caused by edge resonant magnetic perturbations from poloidal and toroidal field coils with small non-axisymmetric displacements [7]. Toroidal asymmetries in the ITER steady-state power flow to the divertors could pose a significant challenge that must be overcome during the machine's operational lifetime starting in 2016 and extending through 2037.

1.3. Transient Particle and Energy Bursts in Fusion Plasmas

While it should be possible to manage the steady-state heat flux in ITER, large transient energy and particle impulses associated with repetitive MHD instabilities known as ELMs [8] are of much greater concern for ITER. When scaled to ITER plasma conditions from current experiments in 1.5 m class

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tokamaks, ELM instabilities are expected to drive impulsive energy bursts reaching 15 MJ with rise times ranging from 300 μs to 500 μs . In ITER the ablation limit of the divertor materials is $58 \text{ MJm}^{-2}\text{s}^{-1/2}$ [9] and the anticipated surface area exposed to these bursts is $\sim 4.7 \text{ m}^2$ so a 15 MJ ELM will deliver an impulsive energy flux of $\sim 67 \text{ MJm}^{-2}\text{s}^{-1/2}$ assuming a toroidally uniform distribution and a 50% radiative fraction along the open field lines. Thus, using the most optimistic assumptions for ITER, it is reasonable to expect that a relatively small number of ELMs (approximately the number occurring in less than a few hundred, full length, high confinement discharges) will rapidly erode the divertors. In addition, observations of edge plasma in the current generation of 1.5 m class tokamaks, using high speed cameras, reveal an array of complex dynamical processes during ELMs such as the formation of rapidly growing coherent filaments that appear to rotate and interact with the divertor as well as the main chamber walls.

While the dynamics and topology of the ELMs in a tokamak are similar in some ways to those of coronal loops, solar flares and coronal mass ejections, they appear to be driven by rather different physical processes [10]. This is not unreasonable to expect since the dynamics of the ELM should be governed to some degree by electromagnetic interactions with the conducting walls of the tokamak and by impurities generated when heat and particle bursts hit plasma facing surfaces. In addition to the violent energy and particle bursts due to ELMs, there is another interesting dynamical process going on in the edge plasma. This involves a relatively constant level of intermittent turbulence that ejects high-density plasma clumps between the ELMs [11–13]. These clumps propagate radially across the magnetic field and hit the main chamber walls but do not appear to carry much energy.

Although the edge of high power tokamak plasmas is an active and topologically complex region, experimental observations of this region appear to be qualitatively consistent with existence of a complex web of magnetic homoclinic tangles defined by intersections of invariant manifolds of the system [14] and with the formation of Hamiltonian chaos [15] in some regions of the edge magnetic field. The following sections will review recent research results relating the properties of these magnetically tangled plasmas and their associated chaotic regions to experimental observations.

2. Hamiltonian Description of Magnetic Field Lines in Tokamak Plasmas

2.1. General Theoretical Approach

A fundamental tool for understanding the structure of the vacuum magnetic fields in the edge of a tokamak is Hamilton–Jacobi theory [16] in which the Hamiltonian $H(\vec{p}, \vec{q})$ is a constant of the motion due to the fact that the phase volume, defined by divergence-free vector fields $\vec{p}, \vec{q}$, does not change during the evolution of the system. Here, $\vec{p} = \partial f / \partial \vec{q}$ is the canonical momentum of the system and $f(\vec{p}, \vec{q})$ is a generating function that describes a canonical transformation of variables $\vec{p}', \vec{q}' \Rightarrow \vec{p}, \vec{q}$ where $\vec{q}$ is an angular variable in the new canonical coordinates. The magnetic field in a tokamak is expressed as:

$$\vec{B} = B_r \vec{e}_r + B_\theta \vec{e}_\theta + B_\phi \vec{e}_\phi = \vec{\nabla} \times \vec{A} \quad (1)$$

where $\vec{A}$ is a gauge-invariant vector potential of the form:

$$\vec{A} = \vec{A}' + \vec{\nabla} F = \chi \vec{\nabla} \theta + \psi \vec{\nabla} \phi \quad (2)$$

and F is an arbitrary function. Here, χ is a toroidal magnetic flux coordinate and ψ is a poloidal magnetic flux coordinate while θ, ϕ are poloidal and toroidal angles respectively [17–21]. When expressed in this form, ψ is effectively a radial variable in the toroidal (ψ, θ, ϕ) coordinate system. Then the canonical form of the magnetic field is found by combining Eqs. (1) and (2) resulting in:

$$\vec{B} = \vec{\nabla} \times \vec{A} = \vec{\nabla} \chi \times \vec{\nabla} \theta + \vec{\nabla} \phi \times \vec{\nabla} \psi \quad (3)$$

This automatically results in a divergence-free magnetic field:

$$\vec{\nabla} \cdot \vec{B} = \vec{\nabla} \cdot (\vec{\nabla} \chi \times \vec{\nabla} \theta + \vec{\nabla} \phi \times \vec{\nabla} \psi) = 0 \quad (4)$$

as required by Maxwell’s equations. The rotational transform of the equilibrium magnetic field, defined as the average rate of change in the poloidal angle with toroidal angle taken over 2π poloidal radians on a flux surface ψ is given by:

$$\iota(\psi) = \left\langle \frac{d\theta}{d\phi} \right\rangle_{\theta(0 \rightarrow 2\pi)} = \frac{\vec{B} \cdot \vec{\nabla} \theta}{\vec{B} \cdot \vec{\nabla} \phi} = \frac{(\vec{\nabla} \phi \times \vec{\nabla} \psi) \cdot \vec{\nabla} \theta}{(\vec{\nabla} \chi \times \vec{\nabla} \theta) \cdot \vec{\nabla} \phi} = \frac{d\psi}{d\chi} \quad (5)$$

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while the rate of change in the toroidal flux coordinate χ with toroidal angle is given by:

$$\frac{d\chi}{d\phi} = \frac{\vec{B} \cdot \vec{\nabla}\chi}{\vec{B} \cdot \vec{\nabla}\phi} = \frac{(\vec{\nabla}\phi \times \vec{\nabla}\psi) \cdot \vec{\nabla}\chi}{(\vec{\nabla}\chi \times \vec{\nabla}\theta) \cdot \vec{\nabla}\phi} = -\frac{d\psi}{d\theta} . \quad (6)$$

Thus, associating ψ with the Hamiltonian H it is seen that the toroidal flux coordinate χ serves as the canonical momentum of the system and the Hamilton–Jacobi equations for the trajectories of the field lines are given by:

$$\frac{d\theta}{d\phi} = \frac{dH}{d\chi} \quad (7)$$

$$\frac{d\chi}{d\phi} = -\frac{dH}{d\theta} . \quad (8)$$

Here, the usual Hamiltonian is recognized in terms of the more familiar p, q canonical coordinates by substituting $\chi \rightarrow p$ and $\theta \rightarrow q$ while associating ϕ with time (t). In a tokamak $2\pi\chi$ is the amount of toroidal magnetic flux enclosed by a surface of constant χ and $2\pi\psi = 2\pi H$ is the poloidal magnetic flux inside a surface of constant H .

Figure 2, a poloidal cross section of the DIII-D tokamak shown in Fig. 1, illustrates the nominally axisymmetric poloidal and toroidal magnetic field coils along with various and non-axisymmetric control coils with respect to the plasma (yellow-orange region in the center of the figure) and the axisymmetric surfaces of constant poloidal magnetic flux $2\pi\psi$. Equations (7) and (8) are generally integrable in the case of an axisymmetric plasma equilibrium but when small non-axisymmetric magnetic fields are present the Hamiltonian becomes an arbitrary function of the toroidal and poloidal angles. In this system a non-axisymmetric symmetry breaking magnetic perturbation can be expressed in terms of a perturbed Hamiltonian $\varepsilon H_1(\chi, \phi, \theta)$ where ε is a small dimensionless perturbation parameter. The total Hamiltonian is then the sum of the axisymmetric part:

$$H_0(\chi) = \int \iota(\chi) d\chi \quad (9)$$

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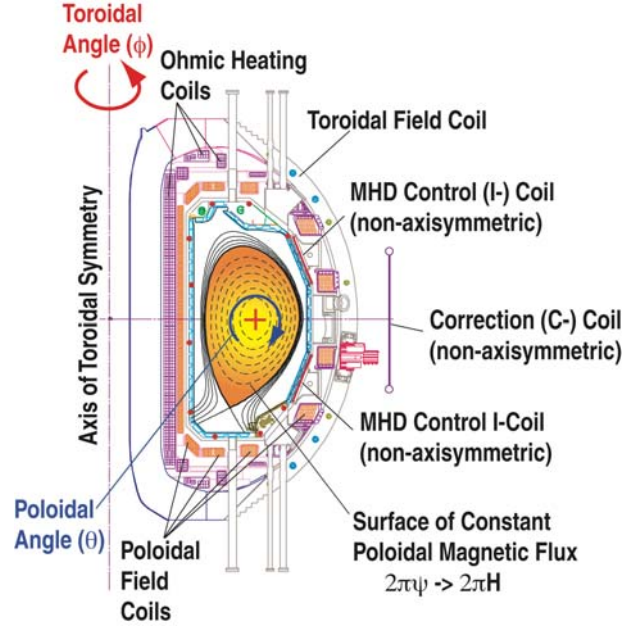


Figure 2. A poloidal cross section of half the DIII-D tokamak shown in Fig. 1 illustrating the locations of various (nominally) axisymmetric and non-axisymmetric magnetic coils along with the toroidal and poloidal angles and the surfaces of constant magnetic poloidal flux indicated with dashed elliptical lines in the hot plasma region (indicated by the yellow-orange region in the center of the figure).

and the non-axisymmetric part:

$$H = H_0(\chi) + \varepsilon H_1(\chi, \phi, \theta) \quad (10)$$

where the perturbed part of the Hamiltonian can be expressed in terms of a Fourier series as:

$$H_1(\chi, \phi, \theta) = \sum_{m,n} H_{m,n}(\chi) \cos(m\theta - n\phi + \chi_{m,n}) \quad (11)$$

Here, m and n are the poloidal and toroidal mode numbers respectively [22].

2.2. *Applications of Hamiltonian Mapping Models to Circular Limited and Poloidally Diverted Tokamaks*

When small non-axisymmetric perturbations are present in the system Eq. (10) is said to be near-integrable and solutions can be found numerically. For example, in the circular limited TCABR tokamak a symplectic Hamiltonian mapping model [23] is used to calculate the perturbed magnetic field from a set of discrete saddle coils wrapped around the outside of the vacuum vessel. Here, the discretized Hamiltonian mapping function is given by:

$$H(\chi, \phi, \theta) = H_0(\chi) + \varepsilon H_1(\chi, \phi, \theta) \sum_{k=-\infty}^{\infty} \delta\left(\phi - k \frac{2\pi}{N_r}\right) \quad (12)$$

where the sequence of δ functions define a discretization of the action-angle variables χ, θ resulting in mapping equations of the form:

$$\chi_{n+1} = \chi_n + \varepsilon f(\chi_{n+1}, \phi_n, \theta_n) \quad , \quad (13)$$

$$\theta_{n+1} = \theta_n + \frac{2\pi}{N_r q(\chi_{n+1})} + \varepsilon g(\chi_{n+1}, \phi_n, \theta_n) \quad , \quad (14)$$

$$\phi_{n+1} = \phi_n + \frac{2\pi}{N_r} \quad , \quad (15)$$

with

$$f = -\frac{\partial H_1}{\partial \theta}, \quad g = \frac{\partial H_1}{\partial \chi} \quad (16)$$

and $\varepsilon (\sim 10^{-4})$ is the current in $N_r = 4$ perturbation coils normalized to the current in the toroidal field coils. Equation (14) is expressed in terms of the safety factor:

$$q(\chi) = 1/\iota(\chi) \quad (17)$$

which is the number of toroidal revolutions needed to complete a single poloidal revolution of the equilibrium magnetic field. Here, resonances in the helical magnetic field, due to m, n Fourier harmonics of the perturbation field occur when:

$$q(\chi_{m,n}) = m/n \quad (18)$$

is a rational number [16].

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Mapping models such as these are similar to the well-known standard map [21,24] which has been used extensively to study stochastic field line trajectories in tokamaks [21,25–28]. In addition, specialized maps, such as the Wobig–Mendonça Map [29,30] and the Tokamap [31], have been formulated to more accurately account for realistic $q(\chi)$ profiles in circular tokamak. A variety of specialized Hamiltonian maps have also been constructed for poloidally diverted tokamak [22,32–35] and so-called “wire” models [36,37] have been used to study magnetic footprints in poloidally-diverted tokamaks. These magnetic footprints determine the poloidal and toroidal distribution of heat and particle flux exhausted from the edge of tokamaks [23,38] and stellarators [39] on to plasma facing material surfaces. Thus, they are of substantial practical importance for designing and operating toroidal magnetic confinement fusion devices. Hamiltonian mapping codes are also an important tool for understanding both the global and fine scale structure of the edge magnetic topology in toroidal confinement systems. This structure can have a fundamental influence on kinetic and fluid transport processes [40] as well as plasma turbulence [41] and MHD stability in the edge [42,43] of the discharge.

3. Hyperbolic Fixed Points and Invariant Manifolds in Tokamak Mapping and Field Line Integration Codes

3.1. General Background

Periodically perturbed continuous vector fields that define the trajectories (flows) of nonlinear dynamical systems such as magnetic field lines in toroidal confinement systems give rise to discrete planar maps referred to as diffeomorphisms [44]. More specifically, if a map is one-to-one and onto (bijective), continuous, and has a continuous, differentiable, inverse then it is referred to as being homeomorphic. Thus, a diffeomorphism is defined as a differentiable homomorphism.

Given a differential equation;

$$\frac{d\xi}{d\phi} = \Lambda(\xi) \quad (19)$$

where $\xi = \xi(\phi)$ is a vector valued function of an independent variable ϕ , the toroidal angle in a tokamak, then a smooth function Λ generates a vector field

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flow $\Omega_\phi : U \rightarrow \mathbf{R}^n$ where $\Omega_\phi(\zeta) = \Omega(\zeta, \phi)$ is a smooth function defined for all ζ in U and ϕ over some interval $I = (a, b) \subseteq \mathbf{R}$ and Ω_ϕ satisfies Eq. (19) since:

$$\frac{d(\Omega(\zeta, \phi))}{d\phi} \Big|_{\phi=\tau} = \Lambda(\Omega(\zeta, \tau)) \quad (20)$$

for all $\zeta \in U$ and $\tau \in I$ [45]. Then a planar map:

$$\Omega(\zeta) = \zeta \quad (21)$$

can be defined as a procedure that images solution curves (trajectories or orbits) of Eq. (19) on to the Poincaré plane. Thus, Ω generates a flow $\Omega : \mathbf{R}^n \rightarrow \mathbf{R}^n$ that can be thought of as the set of all solutions to Eq. (19).

A key requirement for the study of solutions to Eq. (19) is to identify fixed points of the system for which:

$$\Omega^k(\zeta_0) = \zeta_0 \quad (22)$$

where k represents the number of periods or iterations of the map required to return to the fixed point ζ_0 . Thus, a period 2 fixed point:

$$\Omega^2(\zeta_0) = \Omega(\Omega(\zeta_0)) \quad (23)$$

returns to its initial position on the Poincaré plane after two iterations of the map i.e., a composition of the map with itself. Fixed points are the most basic anatomical element of a dynamical system in the sense that the properties of their local (linearized) eigenvalues λ_i uniquely determine the stability properties of all the solutions. For example, a hyperbolic fixed point (sometimes referred to as a saddle point) of Hamiltonian systems has two real eigenvalues that satisfy the condition $|\lambda_s| < 1 < |\lambda_u|$ and the asymptotic stability of the trajectories is determined by invariant sets associated with these eigenvalues. Here, the stable eigenvalue λ_s is associated with an invariant set that contracts trajectories toward its fixed point while the invariant set associated with λ_u expands the trajectories that move away from the fixed point and is said to be unstable. The stable w^s and unstable w^u invariant manifolds of a hyperbolic fixed point ζ_0 generated by the planar map $\Omega(\zeta) = \zeta$ are informally defined as:

$$W^s(\zeta_0) = \{\zeta \mid \lim_{k \rightarrow \infty} \Omega^k(\zeta) = \zeta_0\} \quad (24)$$

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and

$$W^u(\xi_0) = \{\xi \mid \lim_{k \rightarrow \infty} \Omega^{-k}(\xi) = \xi_0\} \quad . \quad (25)$$

An important question in dynamical systems theory is whether the fixed points of the system persists when perturbed. It can be shown using the implicit function theorem that the addition of a small autonomous (time independent) perturbation ε does not destroy the fixed points since the system is continuous and has solutions with continuous derivatives that reside within ε of the unperturbed fixed points although the fixed points may move slightly [46]. Thus, this aspect of the system is stable under small perturbations.

3.2. *Calculations of Homoclinic Tangles and Heteroclinic intersections in Circular Limited Tokamaks*

Edge RMPs were originally proposed as a possible method of controlling the heat and particle exhaust in circular tokamak by producing resonant magnetic islands on rational magnetic surfaces that overlap and create a stochastic boundary layer [47,48]. The first experimental tests of this concept were done in circular tokamaks using modular coils wound around the outside of the vacuum vessel. These coils were used to produce a so-called ergodic magnetic limiter [49,50] or ergodic divertor configuration [1]. These modular coils were also used to carry out the first experimental tests of the resonant helical (island) divertor configuration [51] based on a concept originally proposed by Karger and Lackner [52] that was designed to more efficiently manage the heat and particle flux in circular limited tokamaks. The resonant helical (island) divertor concept, as implemented in a second experiment of this type on the JIPP T-IIU tokamak [53], is shown in Fig. 3. This concept was eventually adopted as an approach for managing the heat and particle flux in stellarators (where it is referred to as the local island divertor configuration) and has proven to be an important factor in obtaining high performance discharges in those systems [54].

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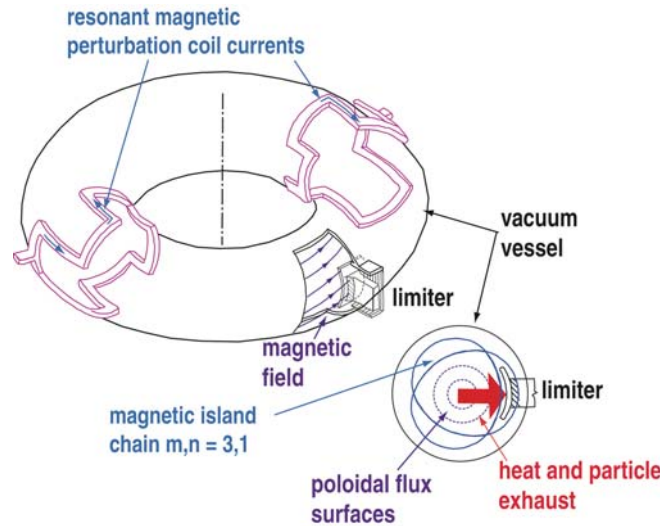


Figure 3. Conceptual implementation of the resonant helical (island) divertor configuration in the JIPP T-IIU tokamak showing the modular RMP coils wrapped around the vacuum vessel and a cutaway view of the magnetic field lines with a limiter inserted into the edge of the plasmas [53]. The poloidal cross section on the right hand side of the figure illustrates the position of the limiter inside the $m,n = 3,1$ resonant magnetic island produced by the currents in the modular coils along with the flow of heat and particles into the limiter on the low magnetic field side of the plasma.

It is noted that the $m,n = 3,1$ magnetic island shown in Fig. 3 is composed of a period $m=3$ hyperbolic fixed point, sometimes referred to as $\times$ -points, where pairs of degenerate w^s and w^u manifolds intersect and $m=3$ elliptic fixed points are located at the center of each island lobe. The field lines inside these helical magnetic islands reside on closed flux surfaces that surround the elliptic fixed points while the field lines outside the invariant manifolds w^s and w^u are confined to the circular (although slightly deformed near the periodic hyperbolic fixed points) poloidal flux surfaces as illustrated in the right-hand part of Fig. 3. As shown below, the structure of the island manifolds depicted in Fig. 3, sometimes referred to as the island separatrix, is an idealization used to illustrate the basic features of the resonant helical (island) divertor concept.

It is often more convenient to model the magnetic field structure in tokamaks using field line integration codes rather than mapping codes since the perturbation fields due to modular coil sets can be easily modeled using current filaments that follow the conductors of the perturbation coils such as those

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shown in Fig. 3. Here, the perturbation fields are calculated using a Biot–Savart algorithm [55] at each step along the integration path and solving a set of field line differential equations:

$$\frac{\partial R}{\partial \phi} = \frac{RB_R}{B_\phi}; \quad \frac{\partial z}{\partial \phi} = \frac{RB_z}{B_\phi} \quad (26)$$

expressed in a cylindrical (R, ϕ, z) coordinate system where B_R, B_ϕ, B_z represent the linear superposition of the axisymmetric equilibrium magnetic field and the non-axisymmetric perturbation fields from all sources in the tokamak of interest. As shown by Bates and Lewis [19], Eq. (26) is Hamiltonian.

When using the field line integration approach and Eq. (26), axisymmetric equilibrium field components can be constructed with N equally spaced circular toroidal field coils, represented by N toroidal magnetic dipoles, and M circular filaments represented by M poloidal magnetic dipoles distributed across the plasma to produce a toroidal current density profile [7]. Various field line integration codes have been used to model the structure of the perturbed magnetic field in circular tokamas [56–59]. These codes typically take small integration steps ($\sim 2\pi/360$) in the direction of the independent variable ϕ and calculate the new R, z position of the field line after each step based on magnetic field components as specified by Eq. (26). Result from these calculations are typically displayed on the Poincaré plane by cutting circular flux surfaces along the outer equatorial plane (typically located at $\theta = 0$ in a toroidal coordinate system) and unfolding them into a rectangular surface where the abscissa represents the poloidal angle θ and the ordinate represents a normalized radial variable (R_N) as shown in Fig. 4. Here, the small dots represent the positions of the field lines as they cross the Poincaré plane following each toroidal transit along their trajectory. Magnetic islands, identified with the $m, n = 1, 1$, $m, n = 2, 1$ and $m, n = 3, 1$ resonant surfaces, are shown in Figure 4 along with examples of invariant manifolds (red curves) superimposed on some of the islands [14]. The invariant manifolds are calculated with a version of the TRIPND code [59] referred to as the TRIP-MAP code [38] that uses an algorithm described by Nusse and Yorke [60] which smoothly resolves segments of the manifolds and with a more refined method described by Hobson [61].

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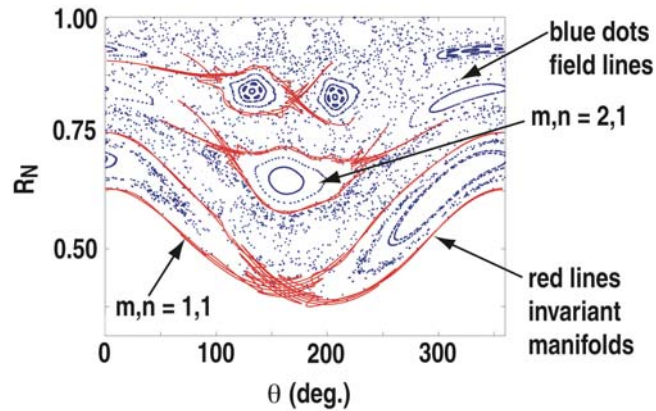


Figure 4. A Poincaré plot of magnetic field line trajectories produced by the TRIPND field line integration for a resonantly perturbed magnetic equilibrium in the Tore Supra tokamak [14,58].

The magnetic structure shown in Fig. 4 results from a statistical model that accounts for multi-millimeter random displacements of the toroidal field coils on the Tore Supra tokamak. Magnetic perturbation sources of this type are referred to as intrinsic topological noise ITN [58]. They result from the fact that tokamaks are built with discrete toroidal and poloidal magnetic field coils which are subject to practical engineering constraints such as tolerance buildups, unbalanced dynamical $\vec{J} \times \vec{B}$ forces due to currents in the plasma and a mutual coupling of the currents in the entire tokamak coil system. In addition, design and construction irregularities, magnetic materials used as part of the heating, diagnostic or control systems and thermal forces on the magnetic components contribute to the ITN spectrum. Thus, it is reasonable to expect that each tokamak has its own unique non-axisymmetric ITN signature resulting from such contributions integrated over the entire ensemble of magnetic components that make up the device. It is also reasonable to expect that the ITN spectrum may change with time as stresses are relieved or as a result of large transient events such as disruptions of the plasma current [62] that can sometime occur in tokamak. On the other hand, so-called “field-errors” result from displacements of individual coils that can be measured [63] and modeled using the Biot–Savart approach discussed above. Modeling of the ITN spectrum in the Tore Supra and TFTR tokamaks, in terms of a Gaussian random normal statistical process, has

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shown that relatively large low m,n modes can be produced that drive resonant magnetic island chains with sizes similar to those associated with field-errors.

A closer look at the invariant manifolds associated with the $m,n = 2,1$ and $m,n = 3,1$ island chains shown in Fig. 4 illustrates the complexity of these structures. Figure 5 shows part of an object generically referred to in dynamical systems theory as a “homoclinic tangle”¹ [60,64]. The homoclinic, self-intersecting, tangle shown in Fig. 5 results from a splitting of the w^s and w^u invariant manifolds associated with a period 2 fixed point (right) and a period 3 fixed point (left) that defines the $m,n = 2,1$ and $3,1$ magnetic island chains respectively [14]. More generally, homoclinic tangles are formed by sets of points that converge to a hyperbolic fixed point ζ_0 (or a fixed point of period k) under the mapping $\Omega^k(\zeta_0) = \zeta_0$ or its inverse $\Omega^{-k}(\zeta_0) = \zeta_0$. Similarly, heteroclinic tangles are formed by sets of points that converge to independent fixed points ζ_0, Σ_0 under the mapping or its inverse.

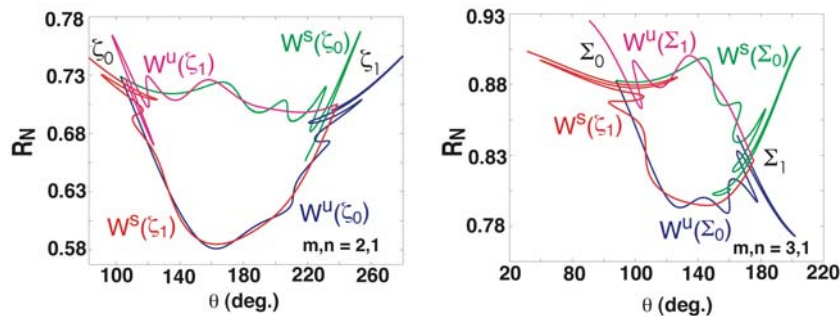


Figure 5. (left) Homoclinic tangle associated with the period 2 fixed point of the $m,n = 2,1$ magnetic island chain and (right) a period 3 fixed point of the $m,n = 3,1$ magnetic island chain for the case shown in Fig. 4 [14].

Intersections of the lobes making up a homoclinic or heteroclinic tangle near fixed points of the system are responsible for stochastic mixing of the field line trajectories. These intersections are permitted when stable tangle manifolds intersect unstable manifolds since this does not violate the uniqueness of

¹ Henri Poincaré discovered these objects while trying to find a solution to a set of equations that describe the motion of a nonlinearly coupled 3-body planetary system and wrote (in Vol 3, Ch. 33, page 389, of *Les Méthodes Nouvelles de la Mécanique Céleste*, Gauthier-Villars, Paris, 1899): “One is struck by the complexity of this figure that I am not even attempting to draw”.

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solutions to the differential equations describing the system. Field lines are also exchanged between neighboring islands through intersections of stable and unstable manifolds associated with fixed points of these islands (heteroclinic intersections) as shown in Fig. 6. The dynamics of the field lines can thus be very complex and have significant implications for the transport and stability properties of the plasma as discussed in Sec. 4.

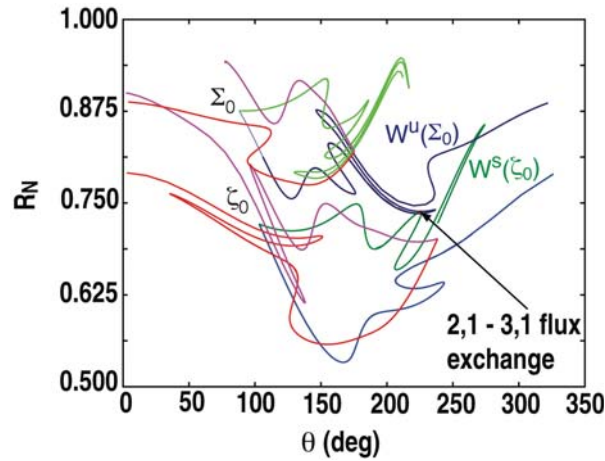


Figure 6. An intersection of a stable invariant manifold associated with the ζ_0 fixed point of the $m,n = 2,1$ magnetic island chain and an unstable invariant manifold associated with the Σ_0 fixed point of the $m,n = 3,1$ magnetic island chain shown in Fig. 4. Magnetic flux exchanged through this type of intersection between islands results in global stochasticity and large scale, non-diffusive, transport of the field lines and the plasma confined to these field lines [14].

3.3. Calculations of Homoclinic Tangles in Poloidally Diverted Tokamaks

Homoclinic tangles are predicted to be a common feature of the primary (nominally axisymmetric) separatrix in diverted tokamak plasma equilibria [38] since relatively small non-axisymmetric magnetic perturbations such as ITN sources, field-errors, helical MHD modes in the core of the plasma, non-axisymmetric corrections and control coils and edge instabilities such as ELMs are typically present at some level. In an ideal, perfectly axisymmetric poloidally diverted tokamak, the stable w^s and unstable w^u manifolds associated with each axisymmetric fixed point are degenerate resulting in an identical

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overlay of one manifold on the other as shown in Fig. 7(a). As first shown by Roeder, Rapoport and Evans [38], vanishing small non-axisymmetric perturbations in these systems produce transverse intersections of w^s and w^u that result in the formation of homoclinic tangles. The intersection points of the manifolds that define a tangle are known as homoclinic intersections or points and it can be shown that an infinite number of such homoclinic points make up the structure of the homoclinic tangle [45]. Additionally, since the system is Hamiltonian and $\vec{\nabla} \cdot \vec{B} = 0$ the magnetic flux passing through each lobe of the tangle is conserved. Thus, as a fixed point is approached either from the stable or unstable direction the lobes get increasingly narrow and stretched out as shown in Fig. 7(b) by the structure of the $w^u(\zeta_0)$ as it approaches the Σ_0 (upper) fixed point along its stable direction.

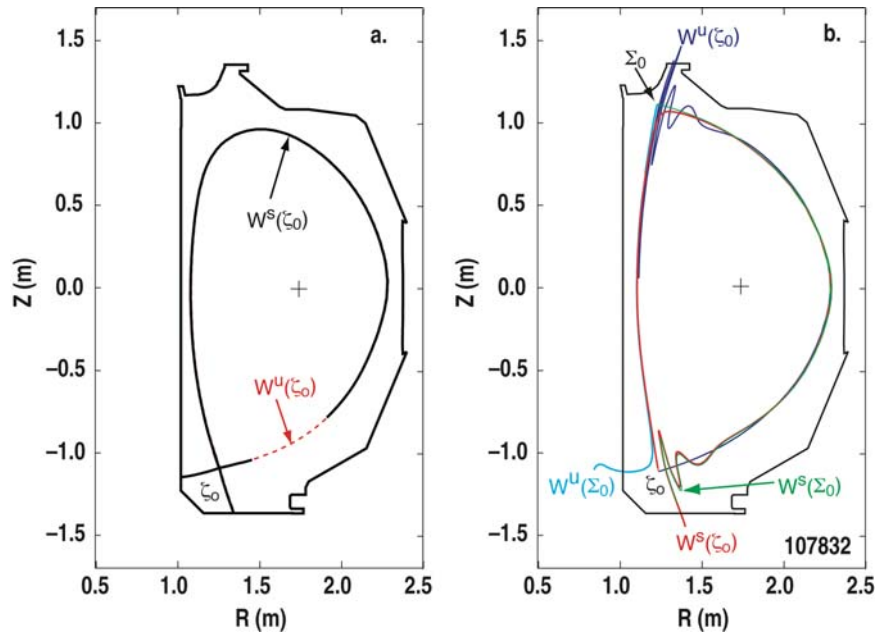


Figure 7. (a) The primary separatrix of an ideal lower single-null poloidally diverted tokamak plasma equilibrium is composed of degenerate stable and unstable invariant manifolds associated with a single hyperbolic fixed point ζ_0 . Here a section of the stable manifold has been removed to reveal the underlying unstable manifold [14]. (b) A balanced double-null poloidally diverted tokamak plasma equilibrium with two independent hyperbolic fixed points ζ_0 and Σ_0 that form a

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complex set of intersecting stable and unstable invariant manifolds when subjected to a non-axisymmetric perturbation due to a field-error correction coil [65].

Although Fig. 7(b) shows only a stable and unstable manifold originating from each fixed point, it should be kept in mind that there are always two stable and two unstable manifolds associated with each hyperbolic fixed point. Here, the tangles that are shown were selected because they have important implications for determining how field line trajectories from the edge of the confined plasma region interact with the material surfaces in the divertors. In other words, they define the boundaries of the magnetic footprints in which field lines from the inner region hit the surfaces and thus allow these nominally closed field lines to open on surfaces in the divertor. This affects the distribution and intensity of the heat and particle flux to the divertors as well as the MHD stability of the edge plasma as discussed in Sec. 4. Figure 7(b) also demonstrates that the invariant manifolds can take on rather complex patterns in which branches of an individual homoclinic tangle shadow branches of another tangle but do not join to form a single heteroclinic tangle. This behavior is controlled by the up-down symmetry of the fixed points and shaping parameters set by the poloidal field coils such as those shown in Fig. 2 [65].

4. Implications of Topologically Tangled Tokamak Discharges on Plasma Confinement and Stability

4.1. General Background

As pointed out above, non-axisymmetric perturbations are always present to some degree in toroidally confined magnetic plasmas and can produce topologically complex structures in the vacuum magnetic fields. These structures are known to produce significant changes in the confinement and stability of the plasma that have direct effects on the power and particle exhaust as well as the overall performance of the discharge. A brief review of such effects is given in this section along with implications for controlling the steady-state heat flux and transient energy pulses as well as particle exhaust rates in ITER and in fusion prototype reactors beyond ITER.

4.2. *Improved Confinement and Transport Barrier Physics in Circular Limited Tokamaks with Stochastic Magnetic Boundary Layers*

The first set of stochastic boundary layer experiments were done in TEXT, a circular tokamak with a ring limiter and a set of modular RMP coils known as the ergodic magnetic limiter (EML) coils [49,50]. The goal of these experiments was to test the idea (originally proposed by Feneberg [47]) that a stochastic layer could be used to cool the edge of the plasma and shield impurities from the core plasma. This process was expected to cause the edge to radiate the heat flux exhausted from the core plasma uniformly to the walls of the vacuum vessel. The physics of energy [66,67] and particle [68] transport was studied in great detail during these experiments and compared to various transport theories [40,69–71]. In particular, it was noted that an anomalously large drop in the plasma density was typically observed during the RMP pulse. This was not predicted by these theories. Detailed measurements of increases in the particle flux around magnetic islands produced by the EML coil was consistent with the existence of an electric field inside magnetic islands $\vec{E}_{isl} \approx 40 \text{ V/cm}$ that resulted in $\vec{E}_{isl} \times \vec{B}$ convective cells near the plasma edge. This convective transport mechanism was found to be large enough to account for the observed drop in plasma density [72,73]. Measurements in the HYBTOK-II tokamak found similar results and support the conclusion that externally driven magnetic structures can significantly alter the electric field in the edge plasma [74]. Recently, this effect has been proposed as a possible mechanism to explain density drops seen during RMP experiments in high confinement (H-mode) plasmas on the DIII-D tokamak [75].

Ergodic magnetic limiter experiments in TEXT also demonstrated that the electron temperature profile could be substantially altered across an edge stochastic layer as shown in Fig. 8. Here, the temperature profile flattens over approximately 75% of the width of the stochastic layer and increases sharply over the remaining 25% of the stochastic layer. This profile has the appearance of a thermal transport barrier located at approximately -23 cm [51] and suggests the possibility that the stochastic boundary layer is creating a type of improved confinement regime. Subsequent experiments, using the modular RMP coils on JIPP T-IIU, also indicated modest improvements in the energy confinement under some conditions [53].

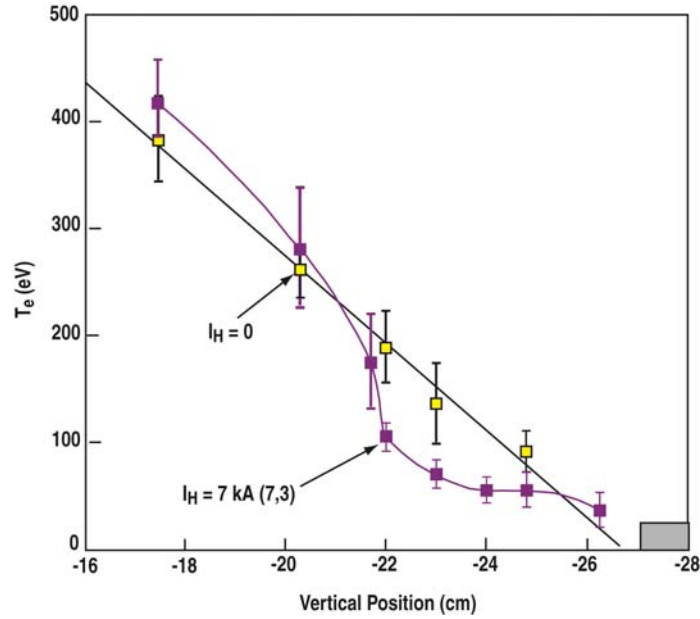


Figure 8. An electron temperature (T_e) profile measured with a Thomson scattering system along a vertical line of sight through the TEXT plasma using the $m,n = 7,3$ EML coil configuration with 7 kA of current compared to the T_e profile in an identical discharge with no EML coil current [51].

Following these initial indications of a new type of improved confinement mode related to the properties of the edge stochastic layer, a series of experiments were done with the ergodic divertor (ED) coils on Tore Supra [76] in order to verify the existence of such a mode [77–79]. In the first experiments with ohmic heating only, it was found that the particle confinement could be increased substantially by positioning the plasma against the inner bumper limiter (the high field side wall), with an edge $q_a = 3.0$, and creating a stochastic layer with the ED coil set [77,78]. Results from the first Ohmically-heated experiment are shown in Fig. 9. When the stochastic layer forms during the ED pulse there is an increase in the line integrated density and the neutral hydrogen particle flux from the limiting surface, shown by the 2D image in Fig. 9(b), drops everywhere within the field of view of the CCD camera as seen in Fig. 9(c). Additionally, it was found that the position of the plasma with respect to the high field and low field side (i.e., good and bad curvature) limiting surfaces

is an important factor for determining whether the confinement increases or is reduced when the stochastic boundary is applied as shown in Fig. 10 [78].

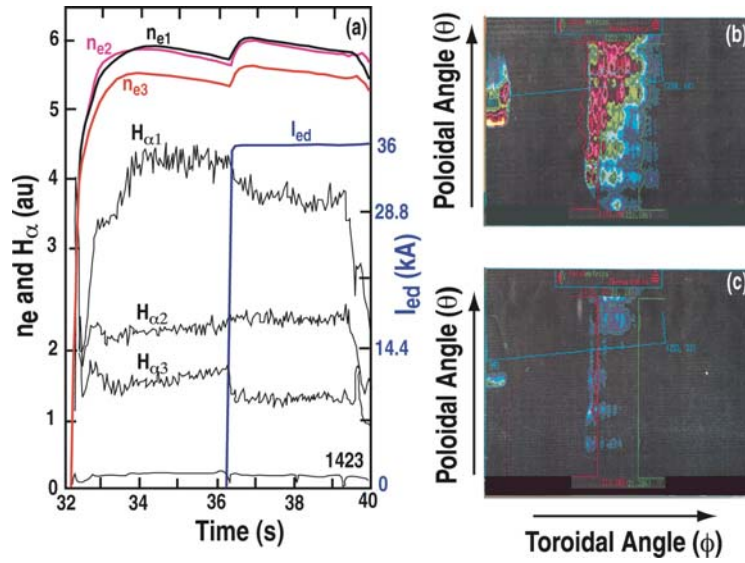


Figure 9. (a) Line integrated density evolution along 3 chords across an Ohmically-heated Tore Supra discharge (#1423) with no external gas fueling after 33 s and the plasma limited on the high field side bumper limiter with a surface safety factor $q_a = 3.0$ [77]. Here, the ergodic divertor coil was energized at 36.1 s with 36 kA of current causing the density to increase while several of the recycling diagnostics ($H_{\alpha1}$ and $H_{\alpha3}$) viewing the high field side limiter showed a decrease in the neutral particle flux from the limiter. (b) A CCD camera image of the H_α emissions with a view normal to the surface of the high field side bumper limiter at 36.0 s showing regions with high levels of neutral recycling (yellow and red) without the ergodic divertor. (c) The same view of the bumper limiter as in (b) but taken at 36.1 s during application of the ergodic divertor.

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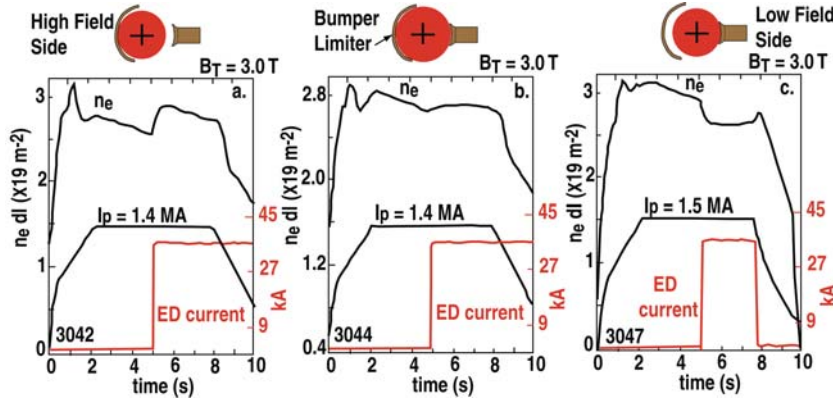


Figure 10. (a) Improved particle confinement with the plasma limited on the high field side bumper limiter during the phase of the discharge when the stochastic layer is applied, (b) a relatively modest increase in the confinement is observed with a stochastic layer when the plasma is limited by both the high field side and the low field side bumper limiters and (c) the confinement is reduced during the stochastic layer phase of the discharge when the plasma is limited on the low field side bumper limiter [78].

These changes in confinement due to geometric differences in the way the plasma is limited on material surfaces suggest that topological effects such as connections between a homoclinic tangle surrounding an island near the surface of the plasma and the wall may be playing a key role in the transport physics. It is known that field lines with short connection lengths to material surfaces can change the electric potential of the plasma, which results in electric fields that affect the bulk motions of the plasma. This is supported by the fact that these improved confinement modes could only be accessed when the safety factor at the plasma surface was within a narrow range around 3.0. Since the peak in the ED perturbation spectrum in Tore Supra was located at $m,n = 18,6$ it is reasonable to assume that a homoclinic tangle associated with the $m,n = 18,6$ magnetic island in combination with a $m,n = 3,1$ field-error island tangle was an important factor in establishing the underlying physics of this effect.

The contrast between improved and degraded confinement regimes in these plasmas was rather dramatic as seen in Fig. 11 where the current in the ED coil set was 45 kA (the maximum current possible). Here, the discharge that transitions to the improved confinement regime, Fig. 11(a), had a slow 2.7 s increase in the stored energy from 140 kJ to 210 kJ with a corresponding linear increase in the plasma pressure normalized by the poloidal magnetic field

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pressure β_p from 0.06 to 0.10. On the other hand, the particle confinement time increased promptly by approximately 55% during the improved confinement phase of this discharge. In the other case shown in Fig. 11(b), with the plasmas limited on the low field side bumper limiter, the stored energy dropped from 275 kJ to 120 kJ on about the same time scale as the drop in density while β_p dropped from 0.12 to 0.05. In this case the particle confinement time drops rapidly following the start of the pulse by $\sim 70\%$ relative to that before the ED field.

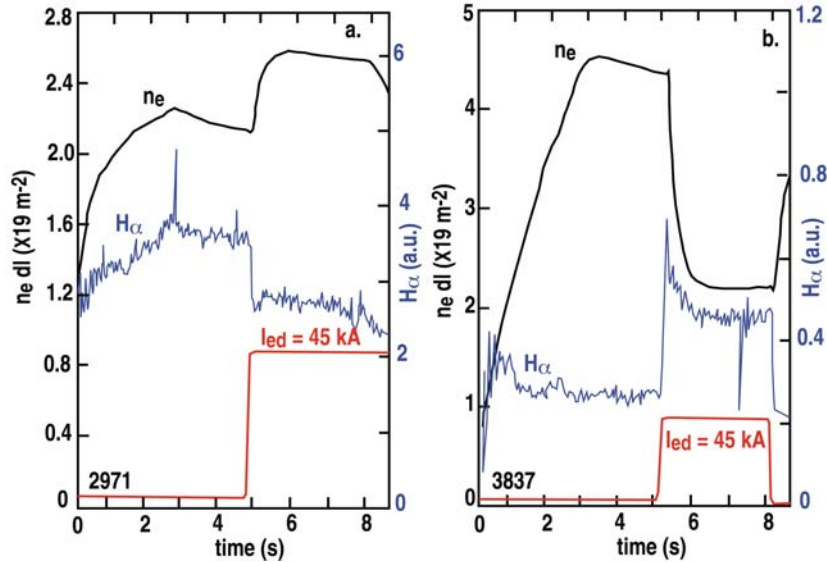


Figure 11. (a) With the plasma limited on the high field side bumper limiter and 45 kA in the ED coil the plasma density increases dramatically while the H_α recycling drops substantially. Note that the rate of change in the density is reduced during the improved confinement phase compared to its value before the ED pulse. (b) With the plasma limited on the low field side bumper limiter the density drops dramatically and the H_α recycling increases indicating a large reduction in the particle confinement time during the ED pulse with the plasma in this configuration [78].

Similar experiments with the plasma limited on the high field side bumper limiter and $q_a = 3.0$ were repeated using lower hybrid current drive (LHCD) with injected power levels of up to 3.3 MW [79]. In these discharges, the volume averaged particle content of the discharge increased from $0.85 \times 10^{19} \text{ m}^{-3}$ during

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the ohmic phase to $\sim 2.1 \times 10^{19} \text{ m}^{-3}$ during the improved confinement phase with the LHCD and an ED coil current of 18 kA. Although the evolution of the density and H_α recycling in these discharges resembles that observed in neutral beam-heated high confinement, H-mode, discharges in poloidally diverted tokamaks such as those required to reach $Q = 10$ in ITER [2], no ELMs were observed and the improvement in the confinement appeared to affect plasma particles more so than the plasma energy [79]. This suggested that an edge stochastic layer may increase the resilience of the discharge to ELM instabilities in high confinement regimes and led to the development of an RMP concept [7] as an option for ELM suppression in H-modes [42,43,75].

These experiments also demonstrated that essentially all of the power exhausted from the core plasma could be radiated uniformly over the plasma facing surfaces due to the onset of an instability, referred to as a MARFE [80–82], which could be maintained for the duration of the combined LHCD and ED pulse. This MARFE resided in the outer 15% of the discharge (approximately the width of the stochastic layer) between the hot core plasma and the high field side bumper limiter [79]. Infrared camera measurements of the temperature distribution on the high field side bumper limiter demonstrated that surface temperature of the limiter was reduced to approximately the same level as that seen during the ohmic phase of the discharge when the ED was turned on during the LHCD pulse and the MARFE was fully formed.

Improvements in the plasma confinement due to the formation of stochastic boundary layer transport barriers have recently been observed in the TEXTOR tokamak using the dynamic ergodic divertor (DED) coil set and neutral beam heating [83]. The TEXTOR results are surprisingly similar to those obtained in Tore Supra including the dependence on the positioning of the plasma with respect to the high field side limiter and the importance of rational q surfaces needed for inducing the transport barrier. Separate DED experiments have also seen the onset of MARFEs [84] that appear to be similar to those obtained in Tore Supra. Reviews of the changes in the electric fields [85] and the transport properties [86] during DED operations in TEXTOR have provided new information on the physics of confinement changes in circular limited plasmas with edge stochastic layers. In addition, work on connecting the structure of homoclinic tangles with heat flux deposition patterns on the high field side bumper limiter has shown a correlation with the DED current [87,88].

4.3. Signatures of Homoclinic Tangles in Poloidally Diverted Tokamaks

While our understanding of confinement and stability physics in circular limited plasmas with edge stochastic layers has progressed substantially due to the results reviewed above, the physics of edge stochastic layers in highly rotating, relatively collisionless, high pressure poloidally diverted H-mode plasmas is substantially unexplored and the few results that do exist are often quite surprising and theoretically challenging [43,75]. As discussed in Sec. 1, managing the steady-state power and transient energy exhaust in the next generation of tokamaks is a critical technological issue that must be solved before a prototype fusion engineering reactor can be designed and built. In addition, the perturbed edge magnetic topology in poloidally diverted tokamaks is considerably more complex than in circular limited tokamaks due to heteroclinic intersections of edge magnetic islands with each other and with homoclinic tangles associated with the primary hyperbolic fixed points that form single and double null diverted configuration. Finally, understanding the self-consistent plasma-magnetic field response to the topology imposed by these complex webs of intersecting homoclinic tangles in high performance H-mode plasmas is far beyond the scope of existing theories.

As discussed above, the first detailed measurements of steady-state toroidal energy deposition asymmetries were made in the ASDEX tokamak [7]. Several years later, measurements made in DIII-D using IR cameras viewing the lower divertor at two toroidally separated positions revealed the complex geometric nature of these toroidal energy exhaust asymmetries. These measurements demonstrated that the heat flux profile at one angle could have a single peak while the heat flux profile at the other toroidal position had two peaks. This resulted in approximately a factor of two asymmetry [89]. Detailed calculations of homoclinic tangles formed by non-axisymmetric field-errors and field-error correction coils produced magnetic footprints that qualitatively matched the splitting of these heat flux profiles [14,90]. It has also been shown that the toroidal phase of the single versus double peak profiles matches the calculated footprint patterns and that these patterns can be controlled with a set of non-axisymmetric coils in DIII-D [90]. A discussion of the properties of homoclinic tangles in DIII-D and their relationship to measurement of divertor heat flux distributions during locked modes and MHD instabilities is given in Ref. 14. It is also pointed out in Ref. 90 that under some conditions the size of the tangles

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can be amplified by the plasma response to the externally imposed magnetic structure.

The first measurements showing that ELMs are composed of non-axisymmetric structures were also made in DIII-D [89]. Here, it was shown that large non-axisymmetric currents flow into and out of the divertor target plates during ELMs. Images of ELMs in MAST, a 1 m class spherical tokamak with $A=1.3$, have explicitly shown the complex 3D topological structure of these instabilities [91]. It has been suggested that the topological structure and evolution of ELM instabilities in tokamak may be related homoclinic tangles driven by ITNs, field-errors or other non-axisymmetric magnetic perturbations that are amplified by the currents flowing in the tangle [92]. It has also been shown in DIII-D that ELMs can be suppressed in ITER relevant plasmas by applying non-axisymmetric magnetic perturbations [42,43,75,93–98].

5. 3D Plasma Modeling of the Plasma Response to Resonant Magnetic Perturbations

A fundamental element needed to understand the physics of tangled edge plasmas in toroidal systems is a theoretical framework that self-consistently couples 3D time dependent plasma equilibria, including the evolution of the plasma pressure and current density, to a set of global stability and confinement theories. This can be thought of as a unified global model of toroidally confined edge plasmas. Although substantial steps have been taken to develop individual parts of such a model an integrated picture is far from being completed.

An example of a 3D equilibrium solver is the VMEC code [99] that imposes perfect magnetic surfaces (no helical islands) on perfectly conducting plasmas while guaranteeing that $\vec{\nabla} \cdot \vec{B} = 0$. The 3D PIES code [100] then uses VMEC equilibria to calculate the topology of resonant flux surfaces, islands and stochastic regions, with increasing plasma pressure. Although these codes were originally developed to model stellarator flux surfaces they have recently been applied to tokamaks with double-null poloidally diverted configurations that conform to stellarator symmetry [101]. Significant progress has also been made in developing 3D transport models both in the area of fluid codes [102–104], that work in moderate to high plasma collisionality regimes, and in the area of gyrokinetic codes [105] that work in low collisionality regimes. Monte Carlo impurity transport codes [106,107] capable of simulating 3D plasmas have also been developed along with a 3D turbulence code [108] and a variety of analytic

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theories [109–114] dealing with various aspects of the transport in stochastic magnetic fields [115].

The final piece needed in an integrated global model edge plasma model is a nonlinear 3D resistive MHD code, for example the NIMROD [116] code, and a perturbed ideal MHD equilibrium code such as CAS3D [117] or IPEC [118]. Thus, there are significant numerical resources available but the task of bringing them all together into a unified model that can be validated against experimental data is still a major step that needs to be completed.

6. Summary and Conclusions

Although magnetized toroidal confinement systems such as stellarators and tokamaks appear to be an attractive route to a steady-state burning fusion plasma regime, a significant number of scientific and technological barriers remain unresolved. For example, a critical technological issue in the next generation of tokamaks is the management of the steady-state power exhaust and transient energy bursts due to edge MHD instabilities known as ELMs. Under ideal conditions, it should be possible to manage the steady-state heat exhaust in ITER but transients due to ELMs must be strongly mitigated or entirely eliminated. In a prototype power producing fusion reactor new solutions will be required to reduce the steady-state heat flux exhaust to the same level expected in ITER [3] and ELMs must be completely eliminated for the duration of the discharge. Techniques based on controlling the edge magnetic topology are being developed to deal with both steady-state heat flux [3,90] and transient energy burst [42,43] issues, but in a more general scientific sense there are fundamental concerns about the number of degrees of topological freedom that are inherent feature of toroidal confinement systems. Since the magnetic field is Hamiltonian and dynamical systems theory predicts that perturbed, divergence-free, vector field flows in toroidal systems result in complex topological structures such as intersecting webs of homoclinic tangles and Hamiltonian chaos, a diverse range of dynamical behaviors can be expected to occur in high power stellarator and tokamak plasmas. Many of these will be driven by the internal dynamics of the plasma interacting with structures produced by field perturbations from external sources that are not well understood from a basic physics point of view.

In circular limited plasmas for example, experiments have shown that the perturbed magnetic field can produce either an increase or a degradation in the

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confinement depending on the specifics of the geometric configuration used and a presumed interaction of homoclinic tangles surrounding a magnetic island chains in the edge plasma with the limiter surface [77,78,83]. In poloidally diverted [14,90] and circular [87,88] tokamaks, measured heat and particle flux profiles hitting plasma-facing surfaces are found to be consistent with magnetic footprints caused by intersections of a homoclinic tangle with the surface. It has also been suggested that violent surface eruptions of the plasma, edge MHD instabilities known as ELMs, may be an expression of the intrinsic topological freedom inherent to the system. Here, it has been proposed that as ELM instabilities drive currents through a preexisting separatrix tangle, caused by field-errors and intrinsic topological noise, during their linear growth phase and these currents amplify the structure of the tangle resulting in a nonlinear, explosive, growth phase during which heat and particles are transported through the lobes of the tangle to plasma facing surfaces [92]. Such a process may be indicative of the potential for the plasma to generate a wide variety of complex structures as the energy density of the system increases, and needs to be understood. It is therefore imperative to develop an interpretive unified model of the edge plasma and to validate the model with experimental data from the current generation of devices.

A variety of numerical models are being developed within the fusion community as discussed in Sec. 5 but have not yet been tested critically against experimental data. Once the individual pieces of these physics models have been validated with experimental data, a unified model can be constructed. A primary constraint on the unified model is that it must preserve the Hamiltonian (symplectic) structure of the magnetic field, which then allows us to make contact with the global topology of the system and to quantify its stability with respect to small perturbations. When viewed from this perspective, a research program of this type can be thought of as an entirely new branch of plasma physics concerned primarily with the global topology of the plasma and may be applicable to a broad range of magnetically confined fusion, solar and astrophysical plasmas.

Acknowledgments

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